



$$q_1 + q_2 + q_3 = Q \quad (1)$$

$$-q_1 + q_4 + q_7 = 0 \quad (2)$$

$$-q_3 - q_4 + q_5 + q_6 = 0 \quad (3)$$

$$-q_6 - q_7 + q_8 = 0 \quad (4)$$

$$q_1 + q_4 - q_3 = 0 \quad (5)$$

$$q_4 + q_6 - q_7 = 0 \quad (6)$$

$$q_2 - q_3 - q_5 = 0 \quad (7)$$

$$q_5 - q_6 - q_8 = 0 \quad (8)$$

$$E = \frac{q_2}{C} \quad (9)$$

$$E = \frac{Q}{C_{AB}} \quad (10)$$

$$q_2 = \frac{8}{15}Q \quad (11)$$

$$C_{AB} = \frac{15}{8}C \quad (12)$$

$$W_0 = EQ \quad (13)$$

$$W_1 = \frac{1}{2}QE \quad (14)$$

$$W_2 = W_0 - W_1 \quad (15)$$